

NGUYEN ANH TUAN

SENIOR SOFTWARE ENGINEER · ASPIRING SOFTWARE ARCHITECT · 11+ YEARS EXPERIENCE

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CAREER OBJECTIVES

Senior Software Engineer with over 11 years of enterprise engineering experience, now making a deliberate transition toward a **Software Architect** role. Throughout my career, I have been involved in designing, developing, and maintaining enterprise-level systems — applying Domain-Driven Design, microservice architecture, and integration patterns across five long-lived products in production. I bring the architectural depth, system-thinking perspective, and technical breadth that a dedicated Architecture role demands: from defining bounded contexts and designing distributed workflows, to integrating complex third-party platforms and delivering scalable solutions for global customers. I am now seeking roles where I can focus on guiding system design decisions, defining technical standards, and shaping the architectural foundations of enterprise platforms.

Core Technologies

Backend: C#, .NET Core, .NET Framework, NodeJS, **Python**, **FastAPI**, OOP, Microservices, Modular Monolith, and Monolithic architectures

Front-End: JavaScript, TypeScript, Angular, ReactJS, HTML5, CSS3

Database: Microsoft SQL Server, SSIS, Azure CosmosDB

Cloud & Infrastructure: Microsoft Azure, App Services, Azure Blob Storage, API-based architecture

DevOps & Messaging: Git CI/CD on Azure DevOps, GitHub Actions, Docker, Docker Compose, Apache Kafka (event streaming) — for automated build, test, and deployment

Integration: RESTful APIs, EDI, JSON-based adapters

Architecture & Design: Domain-Driven Design (DDD), Repository Pattern, Factory Pattern, Singleton Pattern, Strategy Pattern, SOLID principles

AI-Assisted Development: Claude Code, GitHub Copilot AI

INFRASTRUCTURE & PRODUCTION SUPPORT

IIS Administration: Application pool tuning, site/binding configuration, ARR & reverse proxy setup, recycling policies, and performance counters for high-load production sites.

Windows Server: Service configuration, scheduled tasks, event log diagnostics, performance monitoring (PerfMon, Resource Monitor), and remote debugging.

Production Debugging & Troubleshooting: Memory dump analysis (WinDbg / DebugDiag), IIS log forensics, SQL Profiler / Extended Events for query bottlenecks, root-cause analysis for production incidents.

On-call Production Support: Hotfix deployment, incident triage, rollback strategies, and post-mortem reporting for enterprise B2B/EDI customers.

BASIC INFORMATION

Birthday	25/11/1989	Marital Status	Married
Nationality	Vietnamese	Gender	Male

WORK HISTORY

Freelance Software Engineer Independent / Self-Employed

02/2026 – Present

Designing and developing a real-time **Personal Protective Equipment (PPE) Compliance Monitoring System** for industrial workplaces. The system uses IP cameras and motion sensors to automatically detect when workers enter hazard zones without required PPE — safety helmets, protective clothing, and safety footwear — and immediately triggers alerts to safety supervisors to prevent workplace incidents.

PPE Compliance Monitoring System (02/2026 – Present)

- Real-time detection of missing PPE (safety helmet, protective clothing, safety footwear) triggered by motion sensor and IP camera integration
- Event-driven alert pipeline: detection events published to **Apache Kafka** topics → consumed by notification service → instant supervisor alerts
- Angular dashboard for live compliance monitoring, violation history, and safety reporting
- Containerized microservices architecture deployed on **Docker** with Docker Compose orchestration

Architecture & Patterns: Applied **Domain-Driven Design (DDD)** with bounded contexts: Detection (camera/sensor input → PPE analysis), Alert Management (violation events → notification routing), and Compliance Reporting (history, analytics, audit trail). Event-driven architecture using Apache Kafka for decoupled, scalable inter-service communication.

Technologies: Python · FastAPI · Angular (v21+) · TypeScript · Apache Kafka · Docker · Docker Compose · Domain-Driven Design (DDD) · RESTful APIs · WebSocket (real-time feed).

Senior Software Engineer

Truecommerce Vietnam

10/2014 – 01/2026 (11 years 3 months)

Developed and maintained enterprise-level systems using EDI (Electronic Data Interchange) to support automated, secure, and reliable data exchange between internal systems and external partners. Owned long-lived products end-to-end, from architecture and implementation to IIS-based production deployment and on-call support.

Product Development

DiMetrics Product (10/2014 – 12/2019)

Metric Engine — Real-time Business Rule Violation Checking in B2B Systems. Designed and implemented a system to detect real-time violations of business rules within a B2B environment. The solution integrates with multiple subsystems:

- **Filler** – Loads business rule data
- **Insight** – Visualizes data via dashboard reports
- **Notification** – Sends alerts when violations occur
- **Reconciliation** – Implemented automated reconciliation and alerting to proactively detect transaction mismatches between Orders, Shipments, and Invoices, reducing financial risk, preventing billing errors, and improving overall system reliability and trustworthiness

Scheduler Engine: Designed, fixed, and enhanced a scheduler engine that scans data and automatically generates export reports in Excel, PDF, and CSV formats, supporting daily, weekly, monthly, and custom interval-based reporting schedules.

Architecture & Patterns: Led end-to-end system design — decomposed into four subsystems (Filler, Insight, Notification, Reconciliation) with clearly defined responsibilities and data flow. Applied **Factory Pattern** for dynamic rule-handler construction and **Singleton Pattern** for shared scheduler context, ensuring consistent state across long-running console services.

Technologies: *Windows Console Services, HTML, CSS, JavaScript, ASP.NET WebForm, .NET Framework 4.7.2, WCF Services (REST), DevExpress, Microsoft SQL Server (schema design, stored procedures, SSIS, performance optimization), Monolithic architecture.*

DiLabeling Product (01/2020 - 12/2021)

Enterprise labeling software with key features:

- Barcode printing on carton boxes with dynamic label content based on shipment count, product details, customer info, or delivery routes
- Integrated order data from Oracle NetSuite systems via APIs
- Supported flexible printing formats (ZPL, PDF) with industrial printers and high-volume data using HTML carton label printing
- Web-based UI to configure label templates and preview HTML before printing

Architecture & Patterns: Designed the multi-format output architecture — applied **Factory Pattern** to generate ZPL, PDF, and HTML label outputs based on printer type and customer profile, and **Singleton Pattern** for the centralized template cache and printer configuration manager, reducing initialization overhead under high-volume print loads.

Technologies: *WCF Services (REST/SOAP), HTML, CSS, JavaScript, ASP.NET WebForm, DevExpress, Microsoft SQL Server (schema design, stored procedures, SSIS, performance optimization), Monolithic architecture.*

TrueDashboard Automotive Product (01/2022 - 01/2024)

Centralized real-time analytics platform:

- Tracking key B2B metrics: Orders, Order Acknowledgements, Shipments, Invoices
- Interactive chart reports by day, week, month, and year
- Unified dashboard for fast business insights

Architecture & Patterns: Drove the architectural migration from monolith to microservices — defined bounded contexts for Orders, Shipments, Invoices, and Acknowledgements under **Domain-Driven Design (DDD)**, modelling each as an aggregate with its own invariants. Implemented **Repository Pattern** to decouple business logic from data access, enabling cleaner unit testing and long-term maintainability.

Technologies: *Angular, TypeScript, JavaScript, HTML5, CSS3, .NET Core 8.0, ASP.NET Core Web APIs, JSON-based integration, Kendo UI, AG-GRID, ChartJS (Line, Bar, Pie, Stacked Bar), Microsoft SQL Server (schema design, stored procedures, SSIS, ETL, DAX Query, performance optimization), Microservice Architecture.*

Cloud API Adapter | Universal Adapter Products (02/2024 - 01/2026)

Designed and implemented business rules for POST (Create) and GET (Download) of Orders, Shipments, and Invoices:

- Integrated multiple customer platforms (Shopify, WooCommerce, Wolt, Mikral, etc.)
- Validated and transformed JSON payloads before routing to Gateway Connector
- Owned maintenance and enhancement of the Universal Adapter, including an Angular-based drag-and-drop UI for data loading and API-based data forwarding to customers

Architecture & Patterns: Designed the integration architecture — modelled each customer platform (Shopify, WooCommerce, Wolt, Mikral) as an isolated bounded context under **Domain-Driven Design (DDD)**, with its own aggregates and transformation rules. Applied **Repository Pattern** for adapter-level persistence and **Strategy Pattern** for platform-specific JSON validation, enabling extensible onboarding of new platforms without core changes.

Technologies: *Angular, Windows Service, .NET Core 8.0, ASP.NET Core Web APIs, JSON-based integration, Kendo UI, AG-GRID, Microsoft SQL Server (schema design, stored procedures, SSIS, performance optimization), Microservice Architecture.*

Certificate Management & Alert System (2024) — Internal Project

Internal system for centrally managing customer certificate files across integration environments. Continuously monitors certificate validity, automatically detects upcoming or expired customer certificates, and sends email notifications and alerts to relevant stakeholders before expiration — preventing service downtime, avoiding broken customer integrations, and ensuring compliance.

Architecture & Patterns: Applied **Repository Pattern** for certificate metadata storage and access, and **Singleton Pattern** for the centralized monitoring and alert-dispatcher service.

Technologies: *Angular, TypeScript, HTML, CSS, JavaScript, .NET Core 8.0, ASP.NET Core Web APIs, JSON-based integration, Kendo UI, AG-GRID.*

Software Engineer

FPT Telecom

12/2012 - 10/2014 (1 year 10 months)

Technologies: *C#, ASP.NET Web Form / MVC - Framework Three Layer, HTML / CSS / JavaScript / jQuery, MSSQL, JSON, AJAX, LinQ to SQL*

Projects: FPTShop.com.vn | Stock Management | MIS - SAPB1

EDUCATION	Post and Telecommunication Institute of Technology Software Technology 09/2007 – 08/2012 8th Semester Scholarship at school.
LANGUAGES	English Intermediate
SKILLS	Languages & Frameworks: C#, OOP, Python, TypeScript, JavaScript, HTML5, CSS3, .NET Core, .NET Framework, ASP.NET Core Web API, FastAPI, Angular, ReactJS, NodeJS Architecture & Design Leadership: System Design · Solution Architecture · Technical Specification · Architecture Decision Records (ADR) · API Design · Integration Architecture · Scalability Planning · Microservices · Modular Monolith · Domain-Driven Design (DDD) · Repository · Factory · Singleton · Strategy · SOLID · Distributed Systems Data & Integration: MSSQL Server, SSIS, RESTful API, WCF, EDI, JSON, Azure Blob Storage Infrastructure & DevOps: IIS · Windows Server · WinDbg / DebugDiag · Docker · Docker Compose · Apache Kafka · GitHub Actions · Git CI/CD on Azure DevOps · Production Debugging & Troubleshooting Testing: Unit Testing with xUnit, MSTest AI-Assisted Development: Claude Code, GitHub Copilot AI Also familiar with: jQuery, MVC Framework, WCF Service
CERTIFICATIONS	ASP.NET Programming · Viet Chuyen Training Center, 2010 Microsoft Certified Systems Administrator · Nhat Nghe ICT Training Centre, 2011